possibilities rather than one of things or facts." And Jordan thought that "observations not only disturb what has to be measured, they produce it"—he claimed that measuring an electron "compel[s] it *to assume a definite position.*" But if there's no quantum world, as Bohr claimed, then measurements can't compel anything to happen there. And Pauli contradicted Bohr too. Pauli thought that observation introduced "indeterminable effects" that disturbed the systems being observed in uncontrollable ways. Yet observation certainly can't disturb the quantum world if there is no quantum world, as Bohr thought. Pauli may have even contradicted himself. He had dismissed the whole enterprise of talking about what happened when nobody was looking. But if it's meaningless to talk about things before they're observed, how could Pauli say observation disturbed anything at all? And Heisenberg and Jordan clearly contradicted Pauli. They had no qualms about making strong statements about unobserved systems. Thus, the myth that these physicists created a unified Copenhagen interpretation is just that—a myth.

Yet, despite their differences, Bohr, Heisenberg, and the rest of the Göttingen-Copenhagen group had a few things in common. They all agreed that it was pointless to talk about what was "really" happening in the quantum world. Making accurate predictions about the outcomes of measurements was, for them, enough. As Bohr put it years after Solvay, "It is wrong to think that the task of physics is to find out how nature is. Physics concerns what we can say about nature." Quantum physics, then, didn't have to present a coherent or consistent picture of how the world operated—indeed, according to Bohr's complementarity, such a picture was necessarily impossible. It was enough to merely describe measurable features of the world accurately, without talking about what was actually happening. Quantum physics, in short, shouldn't be taken seriously as a theory of the way the world actually is. Instead, quantum physics is a mere tool, an instrument for predicting the outcomes of measurements. Yet, strangely, its unseriousness should be taken very seriously: in claiming their version of quantum physics as a "closed theory," Heisenberg and Born were ruling out the possibility of an explanation of the quantum world, independent of observation, even in principle.

This is where Einstein parted ways with Bohr, Heisenberg, and their ideological compatriots. "The programmatic aim of all physics," according